

Module 3

1. Different 802.11 PHY layer standards & comparison.

Standard	Year	Frequency Band	Technology Used	Max Speed	Key Features
802.11	1997	2.4 GHz	DSSS / FHSS	2 Mbps	First Wi-Fi, very basic
802.11b	1999	2.4 GHz	DSSS	11 Mbps	Good range, high interference
802.11a	1999	5 GHz	OFDM	54 Mbps	Less interference, shorter range
802.11g	2003	2.4 GHz	OFDM	54 Mbps	Backward compatible with 11b
802.11n (Wi-Fi 4)	2009	2.4 / 5 GHz	OFDM + MIMO	Up to 600 Mbps	First MIMO, big speed jump
802.11ac (Wi-Fi 5)	2013	5 GHz	OFDM + 256-QAM + MU-MIMO	Multi-Gbps	Higher QAM, multi-user support
802.11ax (Wi-Fi 6)	2019	2.4 / 5 / 6 GHz	OFDMA + 1024-QAM	~9.6 Gbps	Better efficiency, many devices
802.11be (Wi-Fi 7)	Upcoming	2.4 / 5 / 6 GHz	OFDMA + 4096-QAM	~46 Gbps	Ultra high speed, 320 MHz channels

2. DSSS and FHSS.

DSSS (Direct Sequence Spread Spectrum):

DSSS is a technique where the original data signal is spread over a wider frequency band using a high-speed pseudo-random code (called a chipping sequence). Each data bit is multiplied with this code, producing a higher rate signal. At the receiver, the same code is used to reconstruct the original data.

- Data is combined with a pseudo-random sequence (chips)
- Signal is spread across a wide bandwidth
- Receiver uses the same code to “despread” and recover data

Advantages:

- Good resistance to noise and interference
- Supports higher data rates than FHSS

FHSS (Frequency Hopping Spread Spectrum):

FHSS is a technique where the signal rapidly switches (hops) between different frequency channels in a predefined pseudo-random pattern known to both transmitter and receiver.

- Data is transmitted on one frequency for a short time
- Then it “hops” to another frequency
- Receiver follows the same hopping pattern to receive data

Advantages:

- Reduces effect of narrowband interference
- More secure

3. How do modulation schemes work in the PHY layer? Compare different modulation schemes and their performance across various Wi-Fi standards.

Modulation is the process of converting binary data (0s and 1s) into a wireless signal by changing properties of a carrier wave such as phase and amplitude. In Wi-Fi, after coding and processing, bits are mapped into symbols using modulation techniques. These symbols are then transmitted over the air using OFDM subcarriers. At the receiver, the signal is demodulated to recover the original bits.

Common modulation schemes in Wi-Fi:

- **BPSK:**
1 bit per symbol
Very strong signal, lowest speed
- **QPSK:**
2 bits per symbol
Medium speed, still reliable
- **QAM:**
Uses amplitude + phase
Higher types = more speed
16-QAM → 4 bits
64-QAM → 6 bits
256-QAM → 8 bits
1024-QAM → 10 bits
4096-QAM → 12 bits
Higher QAM = higher speed but needs strong signal

Performance across Wi-Fi standards:

- **802.11a/g:**
Uses up to 64-QAM
Speed up to 54 Mbps

- **802.11n (Wi-Fi 4):**
Same modulation + MIMO
Higher speed
- **802.11ac (Wi-Fi 5):**
Uses 256-QAM
Much higher speed
- **802.11ax (Wi-Fi 6):**
Uses 1024-QAM
Better efficiency
- **802.11be (Wi-Fi 7):**
Uses 4096-QAM
Very high speed

4. What is the significance of OFDM in WLAN? How does it improve performance?

OFDM (Orthogonal Frequency Division Multiplexing):

OFDM is a technique where a Wi-Fi channel is divided into many smaller subcarriers, and data is sent in parallel over these subcarriers.

- Used in all modern Wi-Fi (802.11a and above)
- Replaced older DSSS method
- Forms the base for high-speed Wi-Fi

Advantages:

- Splits data into multiple subcarriers leading to faster transmission
- Handles multipath effectively
- Reduces interference and noise impact
- Uses bandwidth efficiently
- Supports higher data rates

5. How are frequency bands divided for Wi-Fi? Explain different bands and their channels.

Wi-Fi uses unlicensed frequency bands (ISM / UNII) and divides them into channels.

1. 2.4 GHz Band

- Range: 2.400 – 2.4835 GHz
- Channels: 1 to 14 (each 20 MHz wide)
- Non-overlapping channels: 1, 6, 11
- Features:
 - Long range
 - Good wall penetration
 - High interference (Bluetooth, microwave, etc.)

2. 5 GHz Band

- Divided into multiple UNII bands
- Many more channels than 2.4 GHz
- Supports 20 / 40 / 80 / 160 MHz channel width
- Features:
 - Higher speed
 - Less interference
 - Shorter range
 - Some channels require DFS (radar detection)

3. 6 GHz Band (Wi-Fi 6E / 7)

- Large continuous spectrum (~1200 MHz)
- Supports wide channels (up to 320 MHz)
- Features:
 - Very high speed
 - Very low interference
 - Shortest range

6. What is the role of Guard Intervals in WLAN transmission? How does a short Guard Interval improve efficiency?

Guard Interval (GI):

It is a small gap added between OFDM symbols during transmission.

Role of Guard Interval:

- Prevents overlap of signals due to reflections (multipath)
- Avoids Inter-Symbol Interference (ISI)
- Gives time for delayed signals to settle before next symbol

Short Guard Interval:

- Short GI = less gap between symbols
- More symbols can be sent per second
- Increases data rate (about 10–11%)

Condition:

- Works well when signal reflections are low
- In high interference environments, long GI is better

7. Describe the structure of an 802.11 PHY layer frame. What are its key components?

- **Preamble:**
 - Used for synchronization between transmitter and receiver
 - Helps the receiver detect the signal and estimate the channel

- **PHY Header (SIG fields):**
Contains information like data rate, modulation type, and length of the frame
Helps the receiver understand how to decode the incoming data
- **Data Field (PSDU):**
Contains the actual data (MAC frame)
Includes payload, MAC header, and error check (FCS)

8. What is the difference between OFDM and OFDMA?

OFDM	OFDMA
Orthogonal Frequency Division Multiplexing	Orthogonal Frequency Division Multiple Access
Single user at a time	Multiple users at same time
Entire channel used by one device	Channel divided among multiple devices
Parallel data transmission using subcarriers	Subcarriers shared between users
Less efficient for many users	More efficient in dense networks
802.11a/g/n/ac	802.11ax (Wi-Fi 6)
High speed for one user	Supports many users simultaneously

9. What is the difference between MIMO and MU-MIMO?

MIMO	MU-MIMO
Multiple Input Multiple Output	Multi-User Multiple Input Multiple Output
Single user	Multiple users
Sends multiple data streams to one device	Sends data to multiple devices at same time
Increase speed for one user	Improve efficiency for many users
802.11n (Wi-Fi 4)	802.11ac/ax (Wi-Fi 5/6)
One device at a time	Multiple devices simultaneously

10. What are PPDU, PLCP, and PMD in the PHY layer?

- **PPDU (PLCP Protocol Data Unit):**
The complete PHY layer frame transmitted over the air
Includes preamble, header, and data (PSDU)
- **PLCP (Physical Layer Convergence Procedure):**
Converts MAC data into PHY format
Adds preamble and header before transmission
- **PMD (Physical Medium Dependent):**
Handles actual signal transmission and reception
Performs modulation, demodulation, and RF operations

11. What are the types of PPDU? Explain the PPDU frame format across different Wi-Fi generations.

Types of PPDU:

- Legacy PPDU (Non-HT): used in 802.11a/b/g
- HT PPDU: used in 802.11n
- VHT PPDU: used in 802.11ac
- HE PPDU: used in 802.11ax (Wi-Fi 6)
- EHT PPDU: used in 802.11be (Wi-Fi 7)

PPDU frame format (general structure):

All PPDU's mainly have:

- Preamble : synchronization (STF, LTF)
- Header (SIG fields) : rate, modulation, length
- Data field (PSDU) : actual data

Across different Wi-Fi generations:

- **Legacy (a/b/g):**
Simple format with basic preamble and header
- **802.11n (HT):**
Adds HT fields (HT-SIG, HT-LTF)
Supports MIMO and higher speed
- **802.11ac (VHT):**
Adds VHT fields
Supports wider bandwidth and MU-MIMO
- **802.11ax (HE):**
Adds HE-SIG fields
Supports OFDMA and multi-user communication
- **802.11be (EHT):**
Further enhancements for higher speed and multi-link operation

12. How is the data rate calculated ?

Data rate in Wi-Fi depends on multiple PHY layer factors.

Main factors:

- Number of data subcarriers
- Bits per subcarrier (modulation)
- Coding rate
- Number of spatial streams (MIMO)
- Symbol duration (including Guard Interval)

Basic formula:

Data Rate = (Number of subcarriers × Bits per subcarrier × Coding rate × Spatial streams) ÷ Symbol duration